

Correction vs Bear Market Prediction

Research Note for Cypress Spring Analytics

Executive Summary

A common mistake among investors is to use recession indicators to predict market corrections. Historical evidence suggests that corrections (10%-20%) and bear markets (>20%) are driven by different mechanisms and therefore require different predictor sets.

Definitions

Correction: Peak-to-trough drawdown between 10% and 20%.

Bear Market: Peak-to-trough drawdown greater than 20%.

Bear Market Predictors

Target Variable:
 $P(\text{Bear in next 12 months})$

Key Predictors:

1. Yield Curve (10Y Treasury - 3M Treasury)
2. Credit Spread (BAA - Treasury)
3. Conference Board LEI
4. ISM PMI
5. Sahm Rule
6. Market Breadth

These indicators work because major bear markets are usually associated with recession, earnings contraction, tightening credit conditions and deteriorating labor markets.

Correction Predictors

Target Variable:
 $P(\text{Correction in next 6 months})$

Key Predictors:

1. Market Breadth
2. Momentum Divergence
3. VIX Term Structure
4. Sentiment Indicators
5. Valuation (CAPE)
6. Liquidity Measures
7. Change in Credit Spreads

Corrections are often caused by positioning, sentiment, valuation resets, or liquidity shocks rather than recessions.

Mathematical Definitions

Yield Curve

$$YC_t = Y10_t - Y3M_t$$

Credit Spread

$$CS_t = Y_{BAA,t} - Y10_t$$

Sahm Rule

$$Sahm_t = U3_t - \min(U3_{(t-11:t)})$$

Breadth

$$Breadth_t = N(\text{Stocks} > 200DMA) / N(\text{Total Stocks})$$

Momentum

$$M_t = P_t - MA10_t$$

VIX Term Structure

$$VIXTS_t = VIX3M_t - VIX_t$$

CAPE

$$CAPE_t = \text{Price}_t / \text{AvgRealEarnings10Y}$$

Drawdown

$$DD_t = \text{Price}_t / \text{PeakPrice}_t - 1$$

Recommended Models

Bear Market Model:

$$P(\text{Bear}_{12m}) = \text{Logistic}(\text{Credit Spread}, \text{Yield Curve}, \text{LEI}, \text{PMI}, \text{Sahm Rule}, \text{Breadth})$$

Correction Model:

$$P(\text{Correction}_{6m}) = \text{Logistic}(\text{Breadth}, \text{Momentum}, \text{VIX Term Structure}, \text{Sentiment}, \text{CAPE}, \text{Liquidity})$$

Current Assessment (2026)

Bear Market Risk: Low-to-Moderate

Correction Risk: Moderate-to-High

Current conditions resemble 1998-1999 more closely than 2007: high valuations, strong technology leadership, healthy economy, contained credit spreads, and elevated investor optimism.